

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    COUNT(DISTINCT CUSTOMER_ID) AS TOTAL_CUSTOMERS
FROM
    order_t;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    COUNT(DISTINCT CUSTOMER_ID) AS TOTAL_CUSTOMERS
FROM
    order_t
```

Output:

Showing 1 rows

TOTAL_CUSTOMERS
994

What is the distribution of the customers across states?

Solution Query:

```
SELECT
    CT.STATE,
    COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT
FROM
    customer_t AS CT
    JOIN order_t AS OT ON CT.CUSTOMER_ID = OT.CUSTOMER_ID
GROUP BY
    CT.STATE
ORDER BY
    CUSTOMER_COUNT DESC;
```

Output:

Test Cases Run SQL

Result: Passed

Query 1

Query:

```
SELECT CT.STATE,
       COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT
FROM   customer_t AS CT
       JOIN order_t AS OT ON CT.CUSTOMER_ID = OT.CUSTOMER_ID
GROUP BY
       CT.STATE
ORDER BY
       CUSTOMER_COUNT DESC
```

Test Cases Run SQL

Output:

Showing first 10 rows out of 49 rows

state	CUSTOMER_COUNT
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35

Observations

- There are 994 unique customers who have placed orders.
- The top states by customer count are Texas: 97 customers, California: 97 customers, Florida: 86 customers, and New York: 69 customers.
- The distributions of the District of Columbia recorded a lower customer count.

Insights

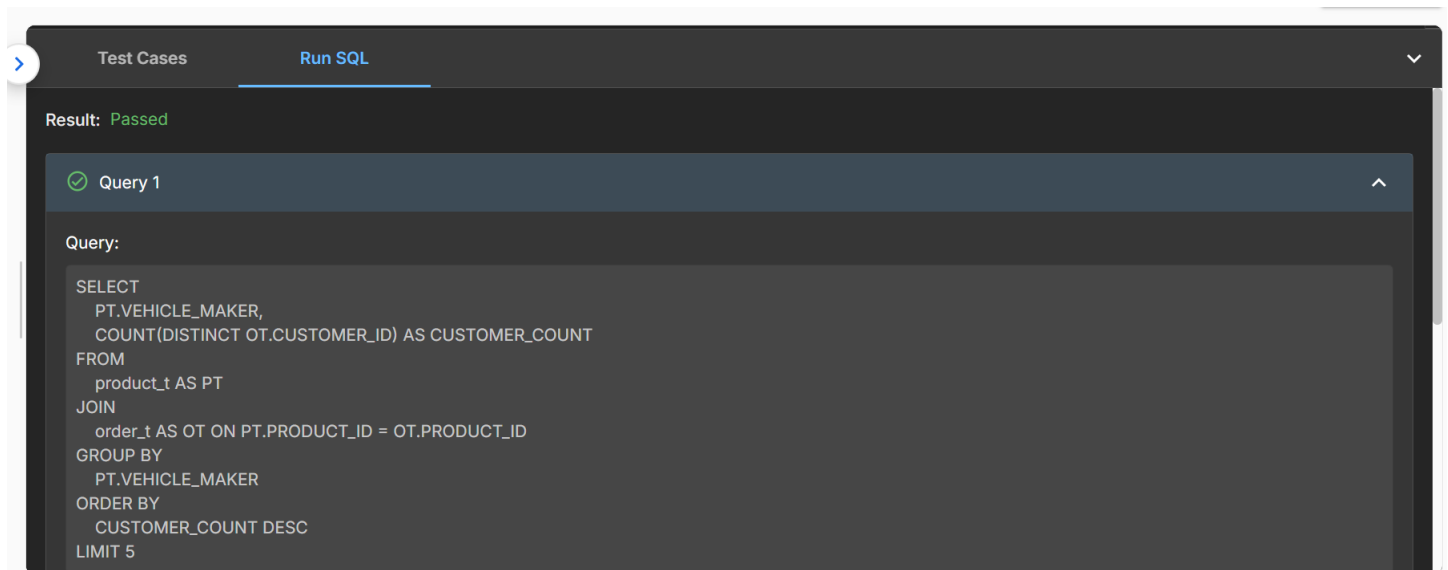
The data demonstrates that Texas and California currently have the highest customer concentration, with Florida and New York following. This finding is valuable for assessing market reach and identifying opportunities for targeted marketing or service enhancements.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

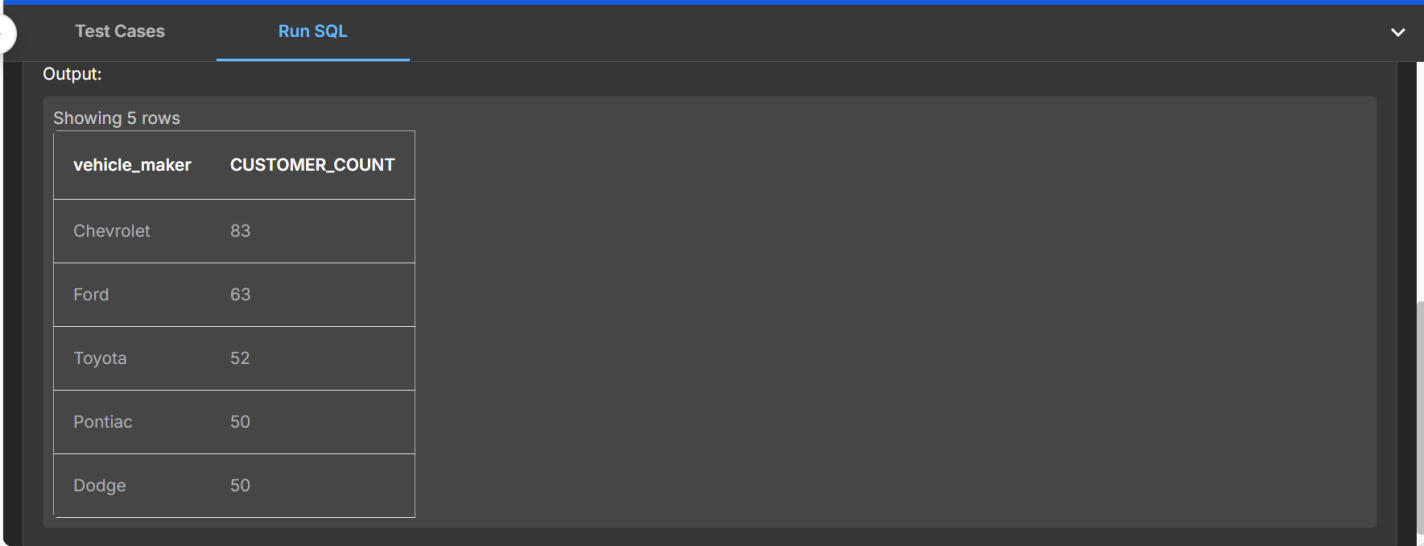
```
SELECT
    PT.VEHICLE_MAKER,
    COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT
FROM
    product_t AS PT
JOIN
    order_t AS OT ON PT.PRODUCT_ID = OT.PRODUCT_ID
GROUP BY
    PT.VEHICLE_MAKER
ORDER BY
    CUSTOMER_COUNT DESC
LIMIT 5;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result is displayed as 'Result: Passed' in green text. Underneath, there is a section for 'Query 1' with a green checkmark icon. The query text is displayed in a monospace font, matching the solution query provided above.

```
SELECT
    PT.VEHICLE_MAKER,
    COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT
FROM
    product_t AS PT
JOIN
    order_t AS OT ON PT.PRODUCT_ID = OT.PRODUCT_ID
GROUP BY
    PT.VEHICLE_MAKER
ORDER BY
    CUSTOMER_COUNT DESC
LIMIT 5
```



The screenshot shows a web application interface with two tabs: 'Test Cases' and 'Run SQL'. The 'Run SQL' tab is active, displaying an 'Output:' section. Below the output header, it says 'Showing 5 rows'. A table is displayed with two columns: 'vehicle_maker' and 'CUSTOMER_COUNT'. The table contains six rows of data, with the first five rows being the top results of the query.

vehicle_maker	CUSTOMER_COUNT
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations

- Chevrolet is the leading vehicle manufacturer, with 83 unique customers.
- Ford ranks second, with 63 unique customers.
- Toyota, Pontiac, and Dodge are also prominent, with 52, 50, and 50 unique customers, respectively.

Insights

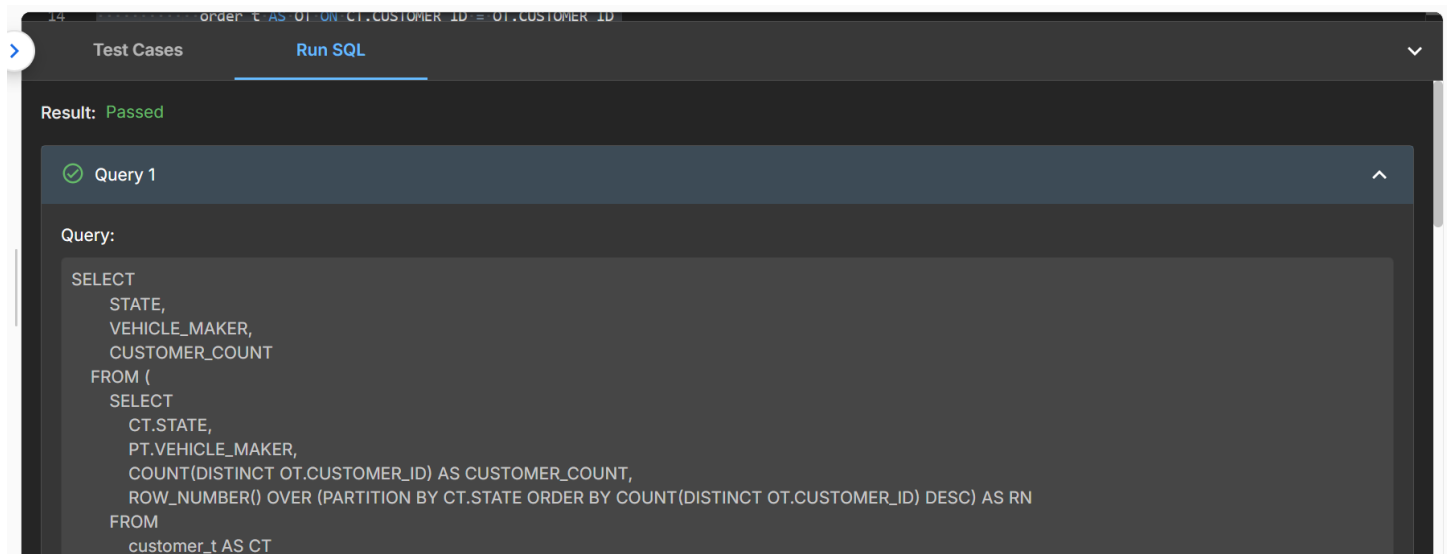
These results suggest a strong preference for American automotive brands, with the Japanese brand Toyota also ranking among the top five. This information is essential for inventory management, marketing strategy development, and the evaluation of potential partnerships with manufacturers.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    STATE,
    VEHICLE_MAKER,
    CUSTOMER_COUNT
FROM (
    SELECT
        CT.STATE,
        PT.VEHICLE_MAKER,
        COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT,
        ROW_NUMBER() OVER (PARTITION BY CT.STATE ORDER BY COUNT(DISTINCT
OT.CUSTOMER_ID) DESC) AS RN
    FROM
        customer_t AS CT
    JOIN
        order_t AS OT ON CT.CUSTOMER_ID = OT.CUSTOMER_ID
    JOIN
        product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
    GROUP BY
        CT.STATE, PT.VEHICLE_MAKER
) AS STATEVEHICLERANK
WHERE
    RN = 1
ORDER BY
    CUSTOMER_COUNT DESC;
```

Output:



The screenshot shows a SQL execution environment with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the status 'Result: Passed' is displayed in green. A section titled 'Query 1' with a green checkmark icon contains the SQL query text. The query is the same as the one provided in the 'Solution Query' block.

```
SELECT
    STATE,
    VEHICLE_MAKER,
    CUSTOMER_COUNT
FROM (
    SELECT
        CT.STATE,
        PT.VEHICLE_MAKER,
        COUNT(DISTINCT OT.CUSTOMER_ID) AS CUSTOMER_COUNT,
        ROW_NUMBER() OVER (PARTITION BY CT.STATE ORDER BY COUNT(DISTINCT OT.CUSTOMER_ID) DESC) AS RN
    FROM
        customer_t AS CT
    JOIN
        order_t AS OT ON CT.CUSTOMER_ID = OT.CUSTOMER_ID
    JOIN
        product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
    GROUP BY
        CT.STATE, PT.VEHICLE_MAKER
) AS STATEVEHICLERANK
WHERE
    RN = 1
ORDER BY
    CUSTOMER_COUNT DESC;
```

```

JOIN
  order_t AS OT ON CT.CUSTOMER_ID = OT.CUSTOMER_ID
JOIN
  product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
GROUP BY
  CT.STATE, PT.VEHICLE_MAKER
) AS STATEVEHICLERANK
WHERE
  RN = 1
ORDER BY
  CUSTOMER_COUNT DESC

```

Test Cases [Run SQL](#)

Output:

Showing first 10 rows out of 49 rows

STATE	VEHICLE_MAKER	CUSTOMER_COUNT
Texas	Chevrolet	9
Florida	Toyota	7
California	Nissan	6
Ohio	Chevrolet	6
Alabama	Dodge	5

Observations

- Chevrolet emerges as the leading vehicle manufacturer in multiple states, including Texas (9 customers), Ohio (6 customers), Colorado (5 customers), Washington (5 customers), the District of Columbia (4 customers), Missouri (4 customers), and Alaska (2 customers).
- Toyota demonstrates significant customer preference in Florida (7 customers), New York (5 customers), Georgia (3 customers), Pennsylvania (3 customers), Oklahoma (2 customers), Oregon (2 customers), Hawaii (1 customer), and Mississippi (1 customer).
- Ford ranks as the most preferred brand in Maryland (5 customers), Virginia (5 customers), and Michigan (3 customers).
- Additional notable preferences are observed for Nissan in California (6 customers) and for Dodge in Alabama (5 customers), Idaho (2 customers), Massachusetts (2 customers), and New Mexico (2 customers).

Insights

State-level analysis reveals unique regional brand loyalties distinct from national patterns. Insights support region-specific marketing, targeted advertising, and inventory management, while highlighting opportunities for market expansion and competitive analysis.



Question 4: Find the overall average rating given by the customers.
What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
  CUSTOMER_ID,
  CUSTOMER_FEEDBACK
FROM
  order_t;
```

Output:

Test CasesRun SQL

Result: Passed

Query 1

Query:

SELECT
 CUSTOMER_ID,
 CUSTOMER_FEEDBACK
FROM
 order_t

Test CasesRun SQL

Output:

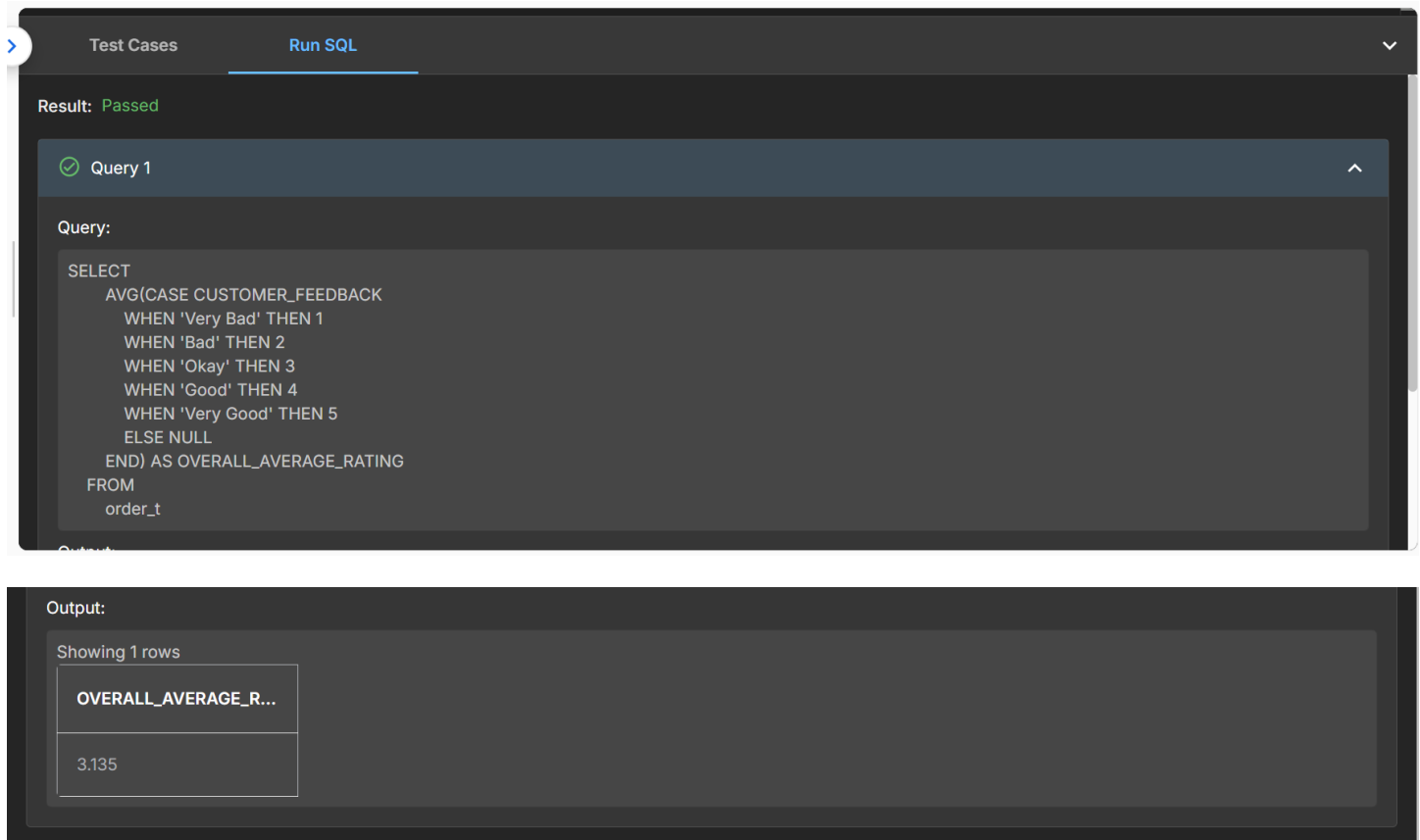
Showing first 10 rows out of 1000 rows

customer_id	customer_feedback
0409-7139	Very Bad
11673-067	Bad
65044-9961	Very Bad
36987-3061	Okay
68151-1471	Okay

SELECT

```
AVG(CASE CUSTOMER_FEEDBACK
  WHEN 'Very Bad' THEN 1
  WHEN 'Bad' THEN 2
  WHEN 'Okay' THEN 3
  WHEN 'Good' THEN 4
  WHEN 'Very Good' THEN 5
  ELSE NULL
END) AS OVERALL_AVERAGE_RATING
FROM
  order_t;
```

Output:



The screenshot shows a SQL execution environment with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the 'Result: Passed' status is displayed in green. A section titled 'Query 1' shows the SQL query being executed. Below the query, the 'Output:' section displays a table with one row and one column.

Result: Passed

Query 1

Query:

```
SELECT
  AVG(CASE CUSTOMER_FEEDBACK
    WHEN 'Very Bad' THEN 1
    WHEN 'Bad' THEN 2
    WHEN 'Okay' THEN 3
    WHEN 'Good' THEN 4
    WHEN 'Very Good' THEN 5
    ELSE NULL
  END) AS OVERALL_AVERAGE_RATING
FROM
  order_t
```

Output:

Showing 1 rows

OVERALL_AVERAGE_R...
3.135

What is the average rating in each quarter?

SELECT

```
  QUARTER_NUMBER,

  ROUND(AVG(CASE CUSTOMER_FEEDBACK
    WHEN 'Very Bad' THEN 1
    WHEN 'Bad' THEN 2
    WHEN 'Okay' THEN 3
```

WHEN 'Good' THEN 4

WHEN 'Very Good' THEN 5

ELSE NULL

END), 3) AS AVERAGE_RATING

FROM

order_t

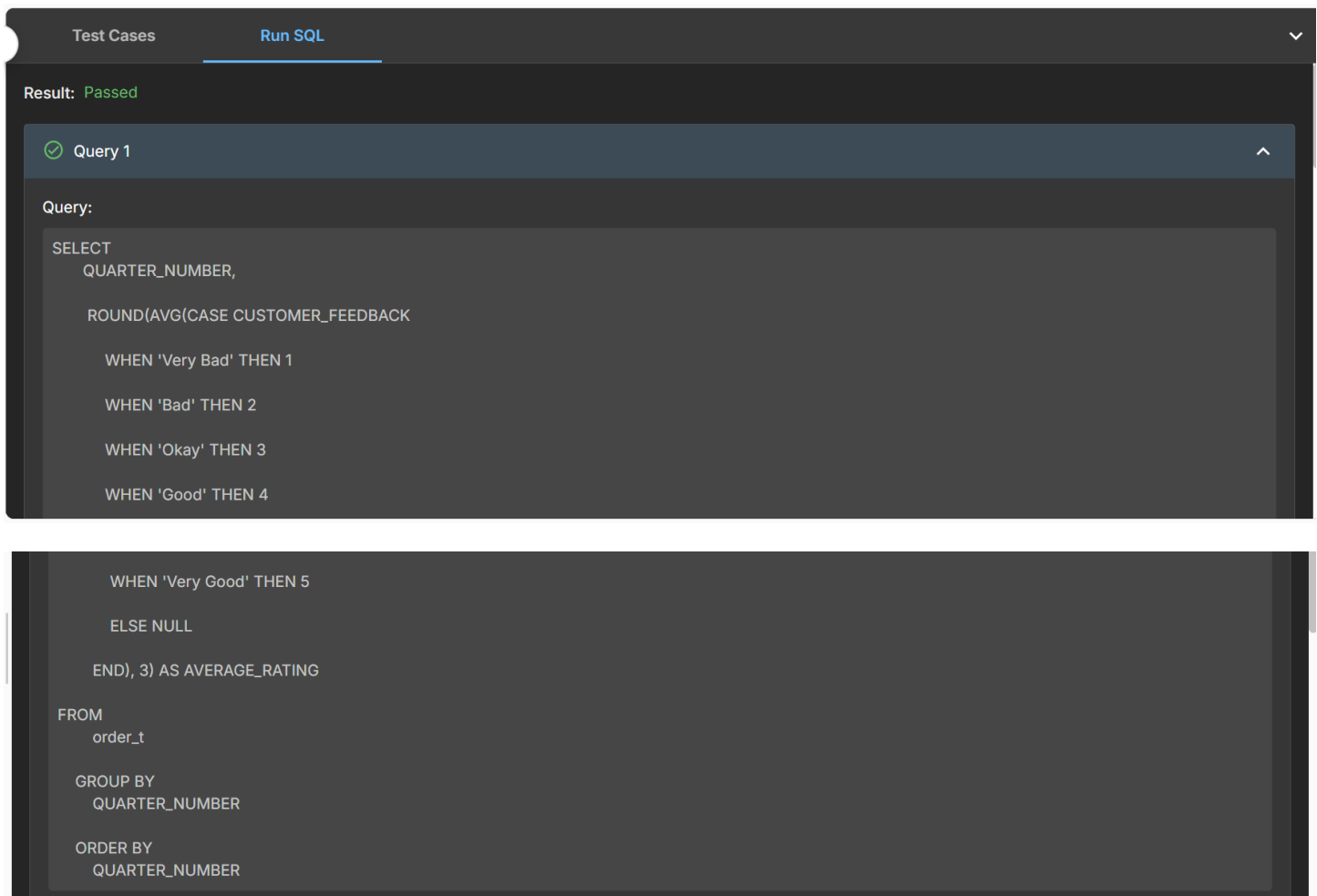
GROUP BY

QUARTER_NUMBER

ORDER BY

QUARTER_NUMBER;

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are two tabs: 'Test Cases' and 'Run SQL', with 'Run SQL' being the active tab. Below the tabs, the result status is 'Result: Passed' in green. A section titled 'Query 1' with a green checkmark icon contains the following SQL query:

```
SELECT
  QUARTER_NUMBER,
  ROUND(AVG(CASE CUSTOMER_FEEDBACK
    WHEN 'Very Bad' THEN 1
    WHEN 'Bad' THEN 2
    WHEN 'Okay' THEN 3
    WHEN 'Good' THEN 4
    WHEN 'Very Good' THEN 5
    ELSE NULL
  END), 3) AS AVERAGE_RATING
FROM
  order_t
GROUP BY
  QUARTER_NUMBER
ORDER BY
  QUARTER_NUMBER
```

Output:

Showing 4 rows

quarter_number	AVERAGE_RATING
1	3.555
2	3.355
3	2.956
4	2.397

Observations

- A rating of 3.13 indicates that, on average, customer feedback is marginally above the 'Okay' benchmark of 3. This result is suboptimal and highlights considerable potential for improvement.
- The most critical insight is the clear and consistent decrease in ratings across the quarters. This suggests systemic issues that are deteriorating over time, rather than isolated incidents.
- The Q4 rating of 2.40 is cause for concern, as it indicates a predominantly negative customer experience by year-end.

Insights

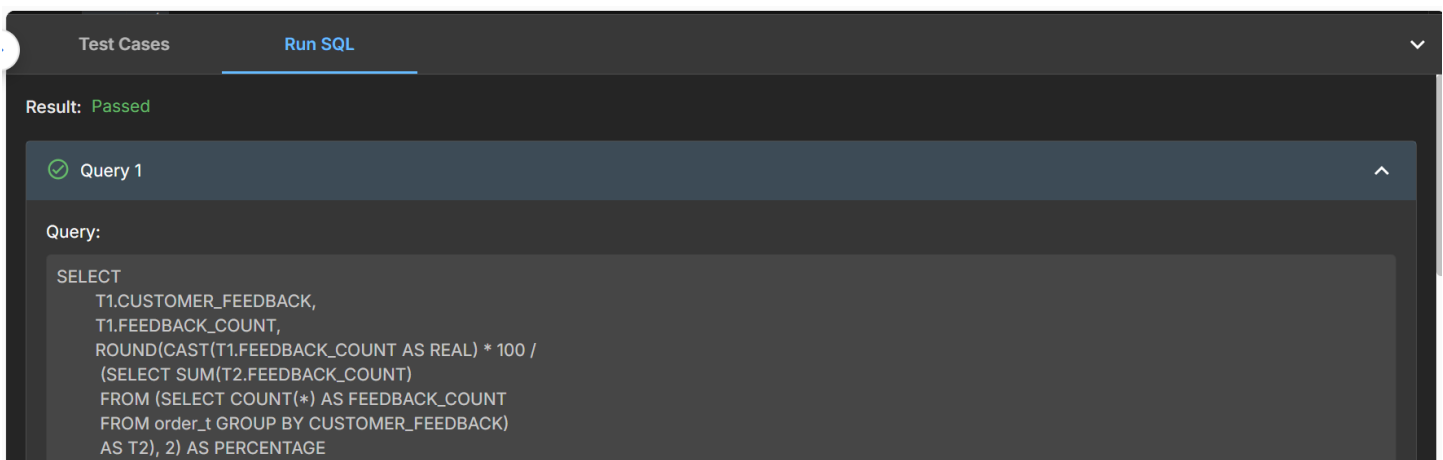
The overall average customer rating is 3.13. However, the quarterly breakdown reveals a downward trend, with the average rating declining from 3.55 in Q1 to 2.40 in Q4. This pattern suggests a significant deterioration in customer experience over the year, with ratings dropping below 'Okay' by Q3 and reaching the 'Bad' category by Q4.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
  T1.CUSTOMER_FEEDBACK,
  T1.FEEDBACK_COUNT,
  ROUND(CAST(T1.FEEDBACK_COUNT AS REAL) * 100 /
    (SELECT SUM(T2.FEEDBACK_COUNT)
     FROM (SELECT COUNT(*) AS FEEDBACK_COUNT
           FROM order_t GROUP BY CUSTOMER_FEEDBACK)
     AS T2), 2) AS PERCENTAGE
FROM
  (
    SELECT
      CUSTOMER_FEEDBACK,
      COUNT(*) AS FEEDBACK_COUNT
    FROM
      order_t
    GROUP BY
      CUSTOMER_FEEDBACK
  ) AS T1
ORDER BY
  PERCENTAGE DESC;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are two tabs: 'Test Cases' and 'Run SQL', with 'Run SQL' being the active tab. Below the tabs, the status 'Result: Passed' is displayed in green. A section titled 'Query 1' with a green checkmark icon contains the SQL query text. The query is the same as the one provided in the 'Solution Query' block.

```
SELECT
  T1.CUSTOMER_FEEDBACK,
  T1.FEEDBACK_COUNT,
  ROUND(CAST(T1.FEEDBACK_COUNT AS REAL) * 100 /
    (SELECT SUM(T2.FEEDBACK_COUNT)
     FROM (SELECT COUNT(*) AS FEEDBACK_COUNT
           FROM order_t GROUP BY CUSTOMER_FEEDBACK)
     AS T2), 2) AS PERCENTAGE
FROM
  (
    SELECT
      CUSTOMER_FEEDBACK,
      COUNT(*) AS FEEDBACK_COUNT
    FROM
      order_t
    GROUP BY
      CUSTOMER_FEEDBACK
  ) AS T1
ORDER BY
  PERCENTAGE DESC;
```

```
FROM
(
  SELECT
    CUSTOMER_FEEDBACK,
    COUNT(*) AS FEEDBACK_COUNT
  FROM
    order_t
  GROUP BY
    CUSTOMER_FEEDBACK
) AS T1
ORDER BY
  PERCENTAGE DESC
```

Output:

Showing 5 rows

CUSTOMER_FEEDBACK	FEEDBACK_COUNT	PERCENTAGE
Very Good	226	22.6
Good	215	21.5
Okay	202	20.2
Bad	182	18.2

Observations

- The distribution of customer feedback is relatively balanced across all categories. Although 'Very Good' is the largest single category.
- The combined negative feedback ('Bad' and 'Very Bad') accounts for 35.7% of all responses.
- This notable proportion of negative feedback, when considered alongside the declining quarterly average ratings, indicates that New-Wheels is experiencing significant customer dissatisfaction.

Insights

The data demonstrate an increasing trend in customer dissatisfaction over time. The average customer rating has steadily declined across the quarters:

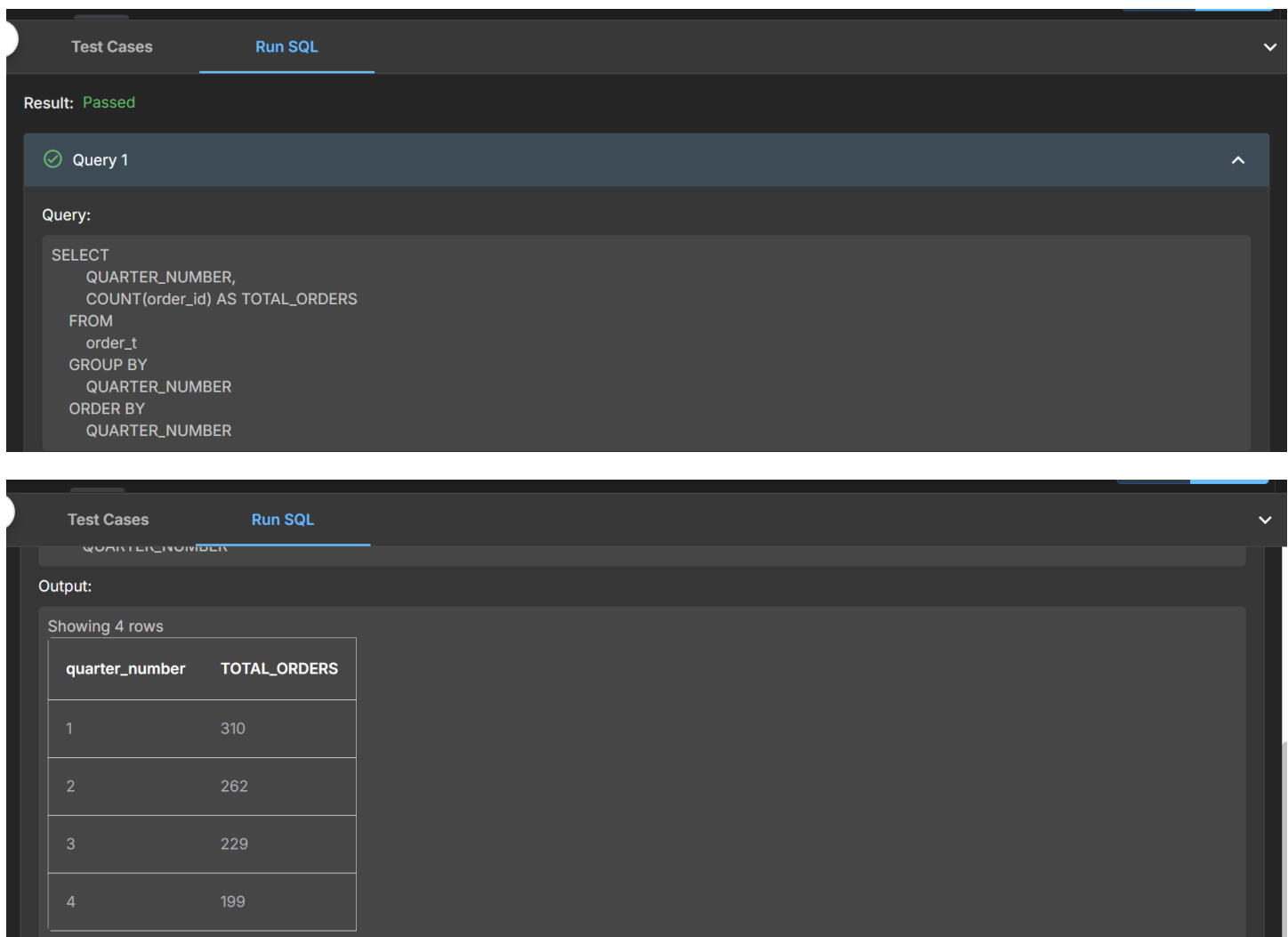
Q1: 3.55; Q2: 3.35; Q3: 2.96; Q4: 2.40. These figures illustrate a clear downward trend, with ratings dropping below 'Okay' by Q3 and entering the 'Bad' category by Q4. This consistent decline indicates a deterioration in customer experience throughout the year.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    QUARTER_NUMBER,
    COUNT(order_id) AS TOTAL_ORDERS
FROM
    order_t
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER;
```

Output:



The screenshot displays a SQL query execution interface. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Passed'. The query is labeled 'Query 1' and is shown in a text area. The output is displayed as a table with 4 rows.

Query:

```
SELECT
    QUARTER_NUMBER,
    COUNT(order_id) AS TOTAL_ORDERS
FROM
    order_t
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER
```

Output:

Showing 4 rows

quarter_number	TOTAL_ORDERS
1	310
2	262
3	229
4	199



Observations

- There is a clear and consistent declining trend in the total number of orders from Q1 to Q4.
- The number of orders has decreased significantly over the year, starting with 310 in Q1 and dropping to 199 in Q4.
- This downward trend in sales directly correlates with the decreasing customer satisfaction observed in the average quarterly ratings.

Insights

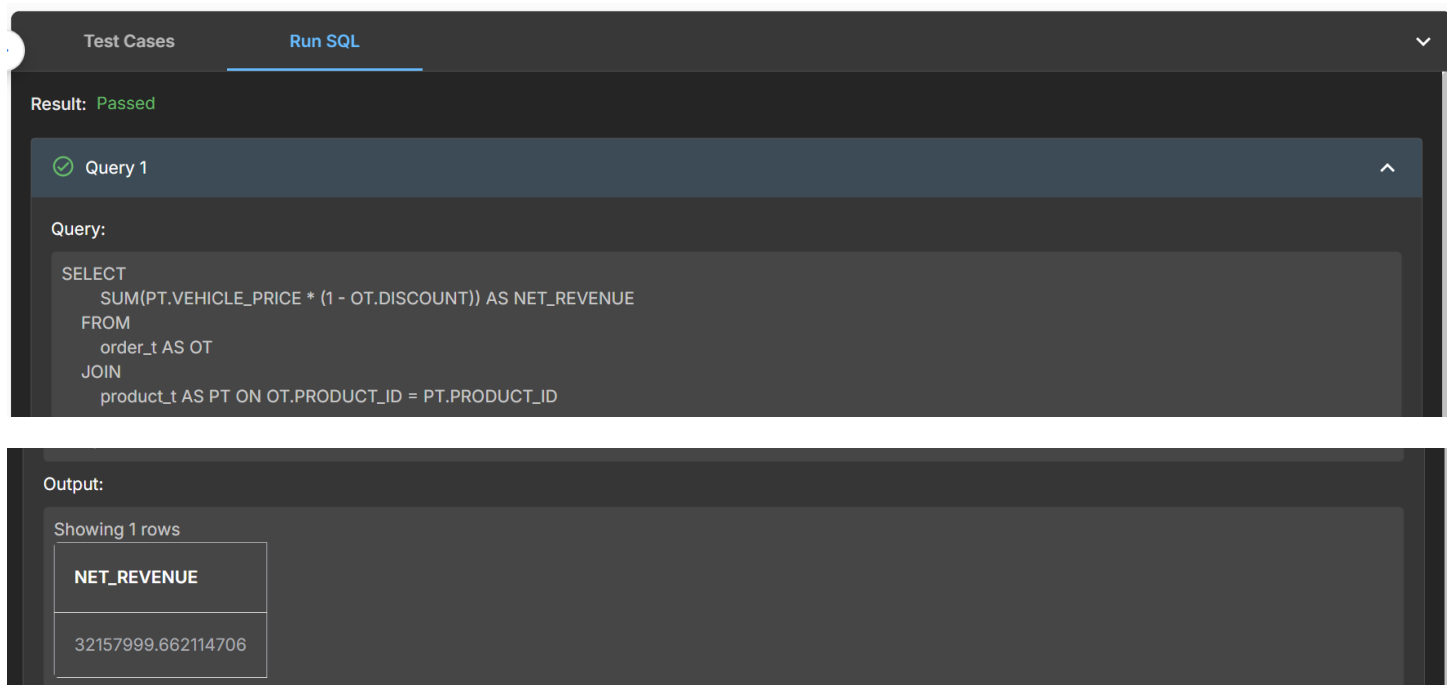
This downward trend in sales directly correlates with the declining customer satisfaction reflected in the average quarterly ratings, suggesting that a worsening customer experience is likely impacting sales volumes. This is a critical concern for New-Wheels.

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    SUM(PT.VEHICLE_PRICE * (1 - OT.DISCOUNT)) AS NET_REVENUE
FROM
    order_t AS OT
JOIN
    product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID;
```

Output:



The screenshot shows a SQL execution environment with two tabs: 'Test Cases' and 'Run SQL'. The 'Run SQL' tab is active, and the result is 'Passed'. Below this, 'Query 1' is displayed with a green checkmark. The query text is the same as provided in the solution. The output section shows 'Showing 1 rows' and a table with one row and one column, 'NET_REVENUE', with the value '32157999.662114706'.

NET_REVENUE
32157999.662114706

What is the quarter-over-quarter % change in net revenue?

```
SELECT
    QUARTER_NUMBER,
    ROUND(SUM(PT.VEHICLE_PRICE * (1 - OT.DISCOUNT)), 6) AS NET_REVENUE
FROM
    order_t AS OT
JOIN
    product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER;
```


Output:

Test Cases
Run SQL

Result: Passed

Query 1

Query:

```

SELECT
    QUARTER_NUMBER,
    ROUND(SUM(PT.VEHICLE_PRICE * (1 - OT.DISCOUNT)), 6) AS NET_REVENUE
FROM
    order_t AS OT
JOIN
    product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER

```

Output:

Showing 4 rows

quarter_number	nNET_REVENUE
1	12100846.3799
2	8584166.776165
3	5956470.396
4	5516516.11005

Observations

- The total net revenue generated by New-Wheels for the entire period is approximately \$32.16 million.
- The pronounced decreases in Q2 and Q3 indicate a rapid deterioration in sales performance.
- New-Wheels has experienced a significant decline in net revenue year-over-year

Insights

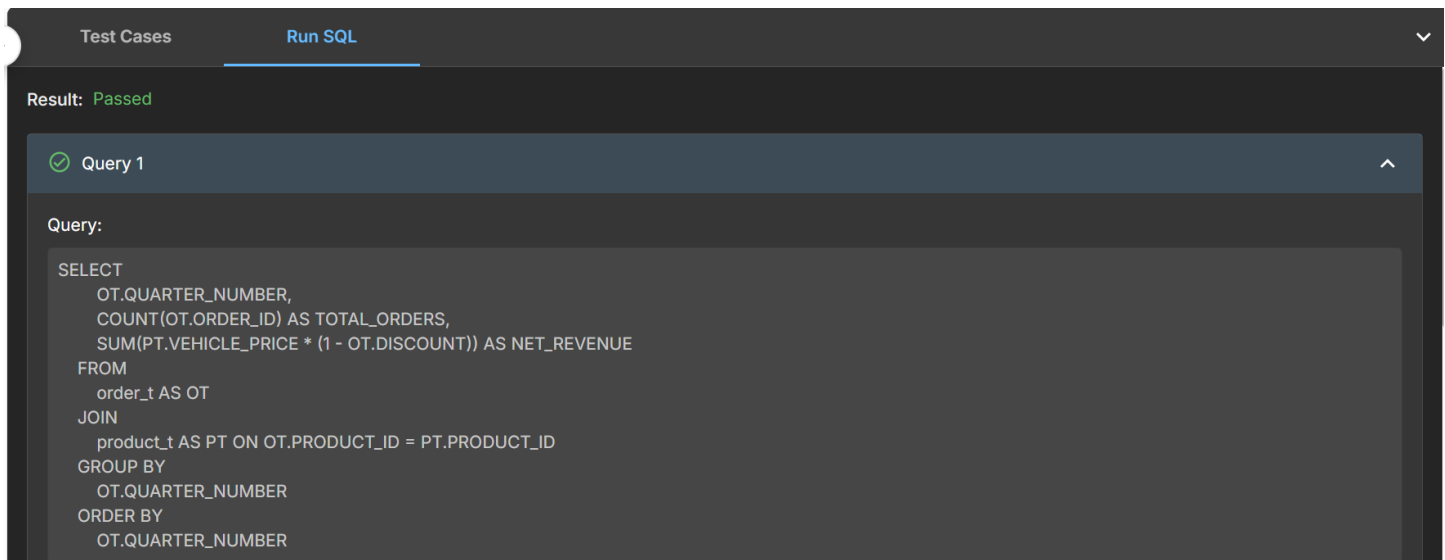
This downward trend in revenue aligns perfectly with the previously observed declines in both the total number of orders and the average customer rating. This strong correlation suggests that diminishing customer satisfaction and reduced order volume are directly impacting the company's financial health.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    OT.QUARTER_NUMBER,
    COUNT(OT.ORDER_ID) AS TOTAL_ORDERS,
    SUM(P.T.VEHICLE_PRICE * (1 - OT.DISCOUNT)) AS NET_REVENUE
FROM
    order_t AS OT
JOIN
    product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
GROUP BY
    OT.QUARTER_NUMBER
ORDER BY
    OT.QUARTER_NUMBER;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are two tabs: 'Test Cases' and 'Run SQL', with 'Run SQL' being the active tab. Below the tabs, the result status is 'Result: Passed' in green text. Underneath, there is a section for 'Query 1' with a green checkmark icon. The query text is displayed in a monospace font, matching the solution query provided in the previous block.

```
SELECT
    OT.QUARTER_NUMBER,
    COUNT(OT.ORDER_ID) AS TOTAL_ORDERS,
    SUM(P.T.VEHICLE_PRICE * (1 - OT.DISCOUNT)) AS NET_REVENUE
FROM
    order_t AS OT
JOIN
    product_t AS PT ON OT.PRODUCT_ID = PT.PRODUCT_ID
GROUP BY
    OT.QUARTER_NUMBER
ORDER BY
    OT.QUARTER_NUMBER
```

Test Cases

Run SQL

OT.QUARTER_NUMBER

Output:

Showing 4 rows

quarter_number	TOTAL_ORDERS	NET_REVENUE
1	310	12100846.379900007
2	262	8584166.776164701
3	229	5956470.395999998
4	199	5516516.110050003

Observations

- This data clearly illustrates a consistent downward trend in both the number of orders and net revenue across all quarters
- Both metrics started strong in Q1 but experienced significant declines, reaching their lowest points in Q4.
- This reinforces earlier observations that declining customer satisfaction directly impacts sales volume and financial performance.

Insights

The business is facing a critical situation, with sales and revenue declining each quarter.

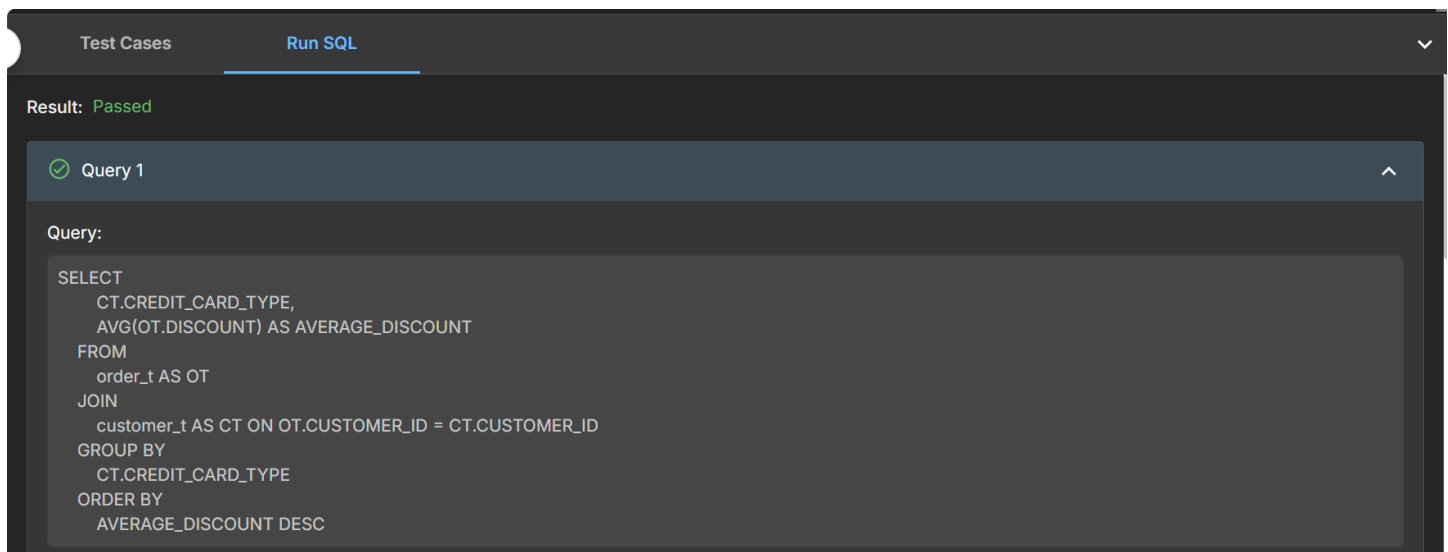
Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

Note: The 'discount' field in the 'order_t' table is stored as a percentage, i.e. 0.6 represents a discount of 0.6%, not 60%.

```
SELECT
    CT.CREDIT_CARD_TYPE,
    AVG(OT.DISCOUNT) AS AVERAGE_DISCOUNT
FROM
    order_t AS OT
JOIN
    customer_t AS CT ON OT.CUSTOMER_ID = CT.CUSTOMER_ID
GROUP BY
    CT.CREDIT_CARD_TYPE
ORDER BY
    AVERAGE_DISCOUNT DESC;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are two tabs: 'Test Cases' and 'Run SQL', with 'Run SQL' being the active tab. Below the tabs, the result status is 'Result: Passed' in green text. Underneath, there is a section for 'Query 1' with a green checkmark icon. The query text is displayed in a monospace font, matching the solution query provided in the text block above.

```
SELECT
    CT.CREDIT_CARD_TYPE,
    AVG(OT.DISCOUNT) AS AVERAGE_DISCOUNT
FROM
    order_t AS OT
JOIN
    customer_t AS CT ON OT.CUSTOMER_ID = CT.CUSTOMER_ID
GROUP BY
    CT.CREDIT_CARD_TYPE
ORDER BY
    AVERAGE_DISCOUNT DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	AVERAGE_DISCOUNT
laser	0.643846153846154
mastercard	0.6294999999999998
maestro	0.6242187499999999
visa-electron	0.623469387755102
china-unionpay	0.6221739130434784

Observations

- The laser credit card type receives the highest average discount at approximately 0.64%.
- Mastercard and Maestro follow with average discounts of around 0.63% and 0.62%, respectively.
- The discounts range down to diners-club-international and solo, which have the lowest average discounts at approximately 0.58%.

Insights

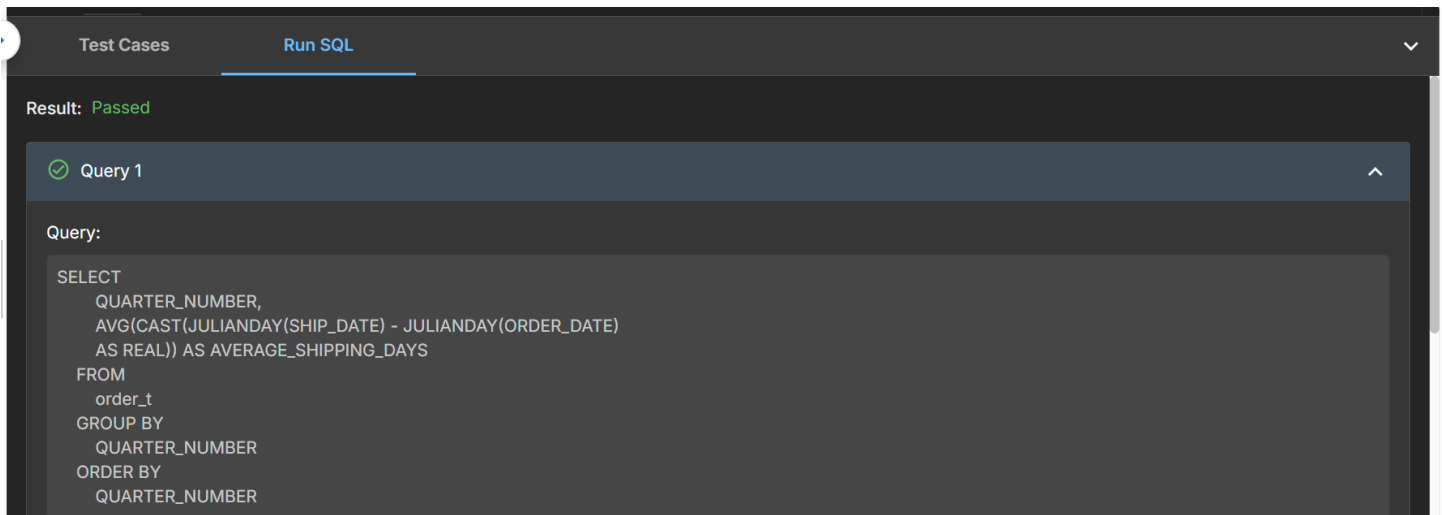
These findings facilitate the identification of credit card types associated with higher discount rates, which may reflect targeted promotional strategies or specific customer segments. Additionally, the results support evaluating the effectiveness of discount campaigns linked to specific card providers.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    QUARTER_NUMBER,
    AVG(CAST(JULIANDAY(SHIP_DATE) - JULIANDAY(ORDER_DATE)
    AS REAL)) AS AVERAGE_SHIPPING_DAYS
FROM
    order_t
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER;
```

Output:



The screenshot shows a SQL execution environment with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Result: Passed'. A section titled 'Query 1' contains the SQL query. The query is as follows:

```
SELECT
    QUARTER_NUMBER,
    AVG(CAST(JULIANDAY(SHIP_DATE) - JULIANDAY(ORDER_DATE)
    AS REAL)) AS AVERAGE_SHIPPING_DAYS
FROM
    order_t
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER
```



The screenshot shows the output of the SQL query. It displays a table with 4 rows. The table has two columns: 'quarter_number' and 'AVERAGE_SHIPPING_DAYS'. The data is as follows:

quarter_number	AVERAGE_SHIPPING_DAYS
1	57.16774193548387
2	71.11068702290076
3	117.75545851528385
4	174.09547738693468

Observations

- This data reveals a critical and alarming trend.
- The average shipping time has drastically increased quarter-over-quarter. Starting from a relatively high 57 days in Q1, it nearly tripled to 174 days by Q4.
- This escalating shipping time is a significant contributor to customer dissatisfaction

Insights

Customers experiencing shipping delays of nearly 6 months (Q4) are highly likely to provide negative feedback, cancel orders, and switch to competitors. These behaviors directly correlate with the observed decline in customer ratings, order volume, and net revenue. New-Wheels should prioritize addressing shipping delays to enhance the customer experience and overall business performance.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
\$125,482,804.37	1000	994	3.13
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$23,496,008.10	199	105 days, 46mins	21.5%

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

- The rise in shipping times from 57 days in Q1 to 174 days in Q4 requires an urgent operational review of supply chain processes, logistics performance, and shipping providers. A comprehensive audit should identify inefficiencies and support improvements. New-Wheels should evaluate alternative shipping partners, renegotiate contracts with accountability measures in place, and enhance logistics systems. Additionally, transparent customer communication regarding shipment status and delays should be implemented to manage expectations effectively
- To address the significant drop in customer satisfaction ratings and elevated levels of negative feedback, it is essential to implement a structured feedback collection and response mechanism. Establishing a service recovery program for dissatisfied customers and executing targeted outreach in brand-loyal states will enhance customer retention and satisfaction while fostering the collection of detailed feedback.
- Utilizing state-level data on preferred vehicle manufacturers enables the development of region-specific marketing initiatives and optimized inventory strategies, ensuring the availability of popular models and improving customer satisfaction. Additionally, it is important to evaluate the impact of credit card-based discount strategies to confirm their effectiveness in promoting sales, while avoiding unnecessary margin reductions, particularly in the context of declining revenue. These actions collectively support improved market performance and profitability.
- Establish a real-time dashboard to track essential performance metrics and set alerts for deviations from targets. Conduct quarterly business reviews to analyze trends and enable swift strategic adjustments. Utilize the existing report format as an initial framework.
- By systematically addressing shipping issues and enhancing the overall customer experience, New-Wheels may reverse the concerning trends of declining sales and customer dissatisfaction, thereby strengthening its financial position.